Current Database Architecture

Schema of Trading Strategy Lab as created by the current ORM and migrations. No live database was opened. Where a comment in an older migration describes tables that `0004_split_live` drops, the head revision is the applied schema.

Analysis date: 2026-10-10.

There are three databases. They do not share foreign keys.

Database	URL setting	Default	Schema source	Bookkeeping
Research	`DATABASE_URL`	`sqlite:///data/genesis.db`	Alembic `backend/alembic`, head `0006_paper_orders`	`alembic_version`
Live	`LIVE_DATABASE_URL`	`sqlite:///data/genesis_live.db`	Alembic `backend/alembic_live`, head `0005_broker_credentials`	`alembic_version` in that file
Market	`MARKET_DATABASE_URL`	`sqlite:///data/market_eod.db`	`MarketBase.metadata.create_all` in `backend/app/market/store.py`	None

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1. Current database technology

Item	Fact
Default platform	SQLite. WAL mode, foreign keys on, writes use `BEGIN IMMEDIATE` (see `backend/app/db/session.py`).
Also accepted	PostgreSQL through SQLAlchemy URL `postgresql+psycopg://...`. `psycopg` is not declared in `backend/pyproject.toml`.
ORM	SQLAlchemy 2. Research models use `Base`. Live models use `LiveBase`. Market models use `MarketBase`.
Primary keys	UUID strings (`String(36)`, Python `uuid4`) or integer `autoincrement`. Not `gen_random_uuid()`.
Money	`Numeric(18, 4)` via `Money` in `backend/app/db/types.py`.
Timestamps	`UTCDateTime`. PostgreSQL stores `timestamptz`. SQLite stores naive UTC and the type decorator restores UTC on read.
JSON	SQLAlchemy `JSON`.
Migrations	Two Alembic configs in `backend/alembic.ini`: default env, and `[live]` (`uv run alembic -n live`).
Market migrations	None. Tables appear on first open.

Research revision chain

`0001_core_sim` → `0002_live_schema` → `0003_journal` → `0004_split_live` → `0005_sim_v3` → `0006_paper_orders`.

`0002_live_schema` created live tables inside the research database. `0004_split_live` drops them. It aborts if any of those tables still has rows. After head, live tables exist only in the live database.

Live revision chain

`0001_live` → `0002_opinions` → `0003_drawdown` → `0004_partial_trades` → `0005_broker_credentials`.

Technology classification

Technology	Status	Evidence
SQLite	Default for all three databases	`backend/app/config.py` defaults
PostgreSQL	Supported for research and live URLs	README and settings. Not the default
Market SQLite	Active, replaceable	Importers rewrite reference data. Reports store a fingerprint instead of auditing this file
pgvector, Redis, Neo4j	Not used	No extension and no client

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2. Current tables

Class:

- **CORE** — written by a current product path.
- **AUDIT** — hash-chained append-only log.
- **VIEW** — not a table.

`strategies` is **NoDelete**: updates are allowed, deletes are not. Every other research and live table below is **AppendOnly**. Market tables are ordinary tables.

Research (`Base`)

Table	ORM	Created	Later changes	Class	Role
`strategies`	`Strategy`	`0001`	—	CORE	Strategy identity. `current_version` and `archived_at` are updated
`ingest_runs`	`IngestRun`	`0001`	—	CORE	One extract attempt
`strategy_versions`	`StrategyVersion`	`0001`	—	CORE	Immutable Markdown snapshot
`reports`	`Report`	`0001`	—	CORE	One Review Crew result
`simulation_runs`	`SimulationRun`	`0001`	`0005` adds data-source and equity columns	CORE	One scenario window
`sim_trades`	`SimTrade`	`0001`	`0005` adds legs, expiry, exit date	CORE	One simulated trade
`audit_events`	`AuditEvent`	`0001`	—	AUDIT	Research hash chain
`trade_reviews`	`TradeReview`	`0003`	`0004` restricts to `trade_kind = 'sim'`	CORE	Reviews of simulated trades
`strategy_improvements`	`StrategyImprovement`	`0003`	—	CORE	Proposed change with evidence
`improvement_decisions`	`ImprovementDecision`	`0003`	—	CORE	Accept, reject, or apply
`paper_orders`	`PaperOrder`	`0006`	—	CORE	One manual paper order from the markets screen
`trade_journal`	—	`0003`, rebuilt in `0004`	—	VIEW	Sim trades joined for `GET /api/journal`

Live (`LiveBase`)

All created in `alembic_live`, not in the research head.

Table	ORM	Creating revision	Later revisions	Class	Role
`deployed_strategies`	`DeployedStrategy`	`0001_live`	—	CORE	Frozen approved strategy
`broker_accounts`	`BrokerAccount`	`0001_live`	—	CORE	Paper or live account
`deployments`	`Deployment`	`0001_live`	—	CORE	One approval of a frozen strategy on an account
`deployment_events`	`DeploymentEvent`	`0001_live`	—	CORE	approved, started, paused, stopped, halted, and later notices
`trade_intents`	`TradeIntent`	`0001_live`	—	CORE	Entry, exit, or protect decision
`orders`	`Order`	`0001_live`	—	CORE	One broker order
`order_events`	`OrderEvent`	`0001_live`	—	CORE	Status and fill progress
`fills`	`Fill`	`0001_live`	—	CORE	One fill
`live_trades`	`LiveTrade`	`0001_live`	`0004` adds `partial`	CORE	Closed or partially closed trade
`broker_charges`	`BrokerCharge`	`0001_live`	—	CORE	Broker-reported charges
`position_snapshots`	`PositionSnapshot`	`0001_live`	`0003` adds `deployment_id`	CORE	Mark-to-market snapshot
`live_trade_reviews`	`LiveTradeReview`	`0001_live`	—	CORE	Reviews of live trades
`live_audit_events`	`LiveAuditEvent`	`0001_live`	—	AUDIT	Live hash chain
`trade_opinions`	`TradeOpinion`	`0002_opinions`	—	CORE	LLM take / reduce / skip
`phantom_trades`	`PhantomTrade`	`0002_opinions`	—	CORE	Lots the veto removed
`capital_events`	`CapitalEvent`	`0003_drawdown`	—	CORE	Deposit or withdrawal
`drawdown_events`	`DrawdownEvent`	`0003_drawdown`	—	CORE	Ladder level change
`broker_credentials`	`BrokerCredential`	`0005_broker_credentials`	—	CORE	Encrypted credential field
`crew_heartbeats`	`CrewHeartbeat`	`0005_broker_credentials`	—	CORE	One row per tick

Market (`MarketBase`)

Table	ORM	Role
`index_bars`	`IndexBar`	Underlying OHLC, India VIX, IV rank
`option_eod`	`OptionEod`	One option contract close
`import_files`	`ImportFile`	SHA-256 of a file already imported
`market_events`	`MarketEvent`	Calendar event

Table	ORM	Role
news_items	NewsItem	Headline

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3. Column-level analysis

Conventions:

- str36 is String(36). strN is String(N). text is Text. int is Integer. bool is Boolean. float is Float. json is JSON. money is Numeric(18, 4). ts is UTCDateTime.
- Nullability follows the ORM. Columns without nullable=True are not null, except where a server default is noted.
- Foreign keys use the naming convention fk_<table>_<column>_<referred table>. ON DELETE is not set, so PostgreSQL uses NO ACTION. SQLite foreign-key enforcement is turned on for the audited databases.
- Check constraints are named ck_<table>_<name>.

strategies

Updates: current_version, archived_at. Deletes: blocked.

Column	Type	Null	Notes
id	str36	no	PK, Python UUID
name	str200	no	
current_version	int	no	Updated when a version is added
created_at	ts	no	
archived_at	ts	yes	Set by archive. The row stays

ingest_runs

Column	Type	Null	Notes
id	str36	no	PK
source_type	str32	no	youtube, instagram, document, notebooklm, markdown
source_url	text	yes	
source_text	text	yes	
source_sha256	str64	yes	
hints	text	yes	
extractor	str32	yes	llm, fixture, or passthrough
model	str100	yes	
raw_output	text	yes	
status	str16	no	
error	text	yes	
actor	str128	no	
request_id	str64	yes	
started_at	ts	no	
finished_at	ts	no	

strategy_versions

Unique (strategy_id, version).

Column	Type	Null	Notes
id	int	no	PK, autoincrement
strategy_id	str36	no	FK strategies.id, indexed
version	int	no	
markdown	text	no	
content_sha256	str64	no	
frontmatter	json	no	Includes rules when present
entry_rules	json	no	List of strings
exit_rules	json	no	
gaps	json	no	
ingest_run_id	str36	yes	FK ingest_runs.id
created_by	str128	no	
created_at	ts	no	

reports

Index ix_reports_strategy_started on (strategy_id, started_at).

Column	Type	Null	Notes
id	str36	no	PK
strategy_id	str36	no	FK strategies.id
strategy_version_id	int	no	FK strategy_versions.id
status	str16	no	
verdict_label	str32	yes	
scenarios_tested	int	yes	
scenarios_passing	int	yes	
inputs	json	no	Provenance snapshot
payload	json	yes	Includes hard_stop_sweep
payload_sha256	str64	yes	
error	text	yes	
actor	str128	no	
request_id	str64	yes	
started_at	ts	no	
finished_at	ts	no	

simulation_runs

Unique (report_id, scenario_index, mode).

Column	Type	Null	Notes
id	int	no	PK
report_id	str36	no	FK reports.id, indexed
scenario_index	int	no	
scenario	json	no	
mode	str16	no	backtest or forward
window_start	str10	yes	
window_end	str10	yes	
bars_in_window	int	no	
trade_count	int	no	
gross_pnl	money	no	
total_costs	money	no	
net_pnl	money	no	
win_rate	float	no	
max_drawdown	money	no	
pass_net	bool	no	
trades_sha256	str64	no	
data_source	str32	yes	Added in 0005. Null on older rows
equity_curve	json	yes	0005
max_drawdown_mtm	money	yes	0005
max_drawdown_days	int	yes	0005
by_year	json	yes	0005
skipped	json	yes	0005

sim_trades

Unique (simulation_run_id, seq). Index ix_sim_trades_report_date on (report_id, trade_date).

Column	Type	Null	Notes
id	int	no	PK
simulation_run_id	int	no	FK simulation_runs.id
report_id	str36	no	FK reports.id
strategy_version_id	int	no	FK strategy_versions.id
seq	int	no	
trade_date	str10	no	
entry_rule	text	no	
entry_conditions	json	no	
reason_text	text	no	
edge	float	no	
market_snapshot	json	no	

Column	Type	Null	Notes
lots	int	no	
credit	money	no	
max_loss	money	no	
notional	money	no	
orders	int	no	
cost_breakdown	json	no	
exit_rule	text	no	
exit_reason	str64	no	
holding_bars	int	no	
target_pnl	money	no	
stop_loss_pnl	money	no	
risk_reward	float	no	
gross	money	no	
costs	money	no	
net	money	no	
pnl_vs_target	money	no	
cost_ratio	float	yes	
outcome_label	str16	no	win, loss, or scratch
equity_after	money	no	
drawdown_after	money	no	
legs	json	yes	0005
expiry	str10	yes	0005
exit_date	str10	yes	0005

trade_reviews

Check sim_trade_only: trade_kind = 'sim' AND sim_trade_id IS NOT NULL.

Column	Type	Null	Notes
id	int	no	PK
trade_kind	str8	no	Must be sim
sim_trade_id	int	yes	FK sim_trades.id, indexed. Check requires it
reviewer_type	str16	no	agent or human
reviewer	str128	no	
rules_version	str16	yes	
followed_plan	bool	yes	
process_verdict	str48	yes	
what_went_well	text	yes	
what_went_wrong	text	yes	

Column	Type	Null	Notes
mistake_tags	json	no	
lesson	text	yes	
improvement_suggestion	text	yes	
suggested_rule_change	json	yes	
confidence	float	yes	
supersedes_review_id	int	yes	FK trade_reviews.id
created_at	ts	no	

strategy_improvements

Column	Type	Null	Notes
id	int	no	PK
strategy_id	str36	no	FK strategies.id, indexed
from_version_id	int	no	FK strategy_versions.id
origin	str16	no	
report_id	str36	yes	FK reports.id
title	str200	no	
rationale	text	no	
proposed_change	json	no	
evidence	json	no	
expected_impact	text	yes	
proposed_by	str128	no	
created_at	ts	no	

improvement_decisions

Column	Type	Null	Notes
id	int	no	PK
improvement_id	int	no	FK strategy_improvements.id, indexed
decision	str16	no	accepted, rejected, or applied
note	text	yes	
resulting_version_id	int	yes	FK strategy_versions.id. Set when applied
decided_by	str128	no	
created_at	ts	no	

paper_orders

Positions are not stored. GET /api/paper/positions nets these rows by contract.

Column	Type	Null	Notes
id	str36	no	PK

Column	Type	Null	Notes
underlying	str32	no	Upper-cased symbol
instrument	str8	no	EQ, FUT, CE, or PE
strike	float	yes	Options only
expiry	str10	yes	YYYY-MM-DD
side	str4	no	BUY or SELL
lots	int	no	Shares for EQ
lot_size	int	no	1 for EQ, else from LOT_SIZES
price	float	no	Price the user typed
charges	json	no	Cost breakdown plus total
strategy_id	str36	yes	FK strategies.id
note	text	yes	
actor	str128	no	
created_at	ts	no	

audit_events and live_audit_events

Same columns. Unique prev_hash and unique hash. Index on (entity_type, entity_id) (ix_audit_events_entity, ix_live_audit_events_entity).

Column	Type	Null	Notes
id	int	no	PK
occurred_at	ts	no	
actor	str128	no	
request_id	str64	yes	
action	str64	no	
entity_type	str64	no	
entity_id	str64	no	
details	json	no	
prev_hash	str64	no	
hash	str64	no	

deployed_strategies

No foreign key to the research database. strategy_id and strategy_version_id are copies.

Column	Type	Null	Notes
id	str36	no	PK
strategy_id	str36	no	Indexed copy
strategy_version_id	int	no	Copy
version	int	no	
name	str200	no	

Column	Type	Null	Notes
markdown	text	no	Frozen text
content_sha256	str64	no	
rules	json	no	Frozen rules
rules_sha256	str64	no	
report_id	str36	yes	Copy
verdict_label	str32	yes	
approved_by	str128	no	
approved_at	ts	no	

broker_accounts

Column	Type	Null	Notes
id	str36	no	PK
broker	str32	no	paper or sandbox in this tree
account_ref_masked	str64	no	
mode	str16	no	paper or live
default_algo_id	str64	yes	
created_at	ts	no	

deployments

Column	Type	Null	Notes
id	str36	no	PK
deployed_strategy_id	str36	no	FK deployed_strategies.id, indexed
strategy_id	str36	no	Indexed copy
strategy_version_id	int	no	Copy
broker_account_id	str36	no	FK broker_accounts.id
mode	str16	no	
sebi_algo_id	str64	yes	
risk_limits	json	no	Capital, lots, drawdown, policy
approved_by	str128	yes	
created_at	ts	no	

deployment_events

Column	Type	Null	Notes
id	int	no	PK
deployment_id	str36	no	FK deployments.id, indexed
event	str32	no	Latest event is the deployment state
reason	text	yes	

Column	Type	Null	Notes
actor	str128	no	
created_at	ts	no	

trade_intents

Column	Type	Null	Notes
id	str36	no	PK
deployment_id	str36	no	FK deployments.id, indexed
deployed_strategy_id	str36	no	FK deployed_strategies.id
strategy_version_id	int	no	
action	str16	no	
legs	json	no	
entry_rule	text	yes	
entry_conditions	json	no	
reason_text	text	no	
market_snapshot	json	no	
thesis_ref	str128	yes	
target_pnl	money	yes	
stop_loss_pnl	money	yes	
max_loss	money	yes	
risk_reward	float	yes	
confidence	float	yes	
risk_checks	json	no	
decided_by	str128	no	
model	str100	yes	Set when an LLM note was attached
approval	str16	no	auto, blocked, or vetoed
approved_by	str128	yes	
decided_at	ts	no	

orders

client_order_id is unique.

Column	Type	Null	Notes
id	str36	no	PK
trade_intent_id	str36	no	FK trade_intents.id, indexed
client_order_id	str64	no	Unique
broker_order_id	str64	yes	
exchange_order_id	str64	yes	
sebi_algo_id	str64	yes	

Column	Type	Null	Notes
exchange	str16	no	
tradingsymbol	str64	no	
expiry	str10	yes	
strike	money	yes	
option_type	str4	yes	
side	str4	no	
quantity	int	no	
order_type	str16	no	
product	str16	no	
limit_price	money	yes	
trigger_price	money	yes	Stop-limit trigger
placed_at	ts	no	

order_events

Column	Type	Null	Notes
id	int	no	PK
order_id	str36	no	FK orders.id, indexed
status	str32	no	
filled_qty	int	no	Default 0
pending_qty	int	no	Default 0
avg_price	money	yes	
message	text	yes	
raw	json	no	
event_at	ts	no	
received_at	ts	no	

fills

Unique (order_id, broker_trade_id).

Column	Type	Null	Notes
id	int	no	PK
order_id	str36	no	FK orders.id, indexed
broker_trade_id	str64	no	
quantity	int	no	
price	money	no	
est_cost_breakdown	json	no	
filled_at	ts	no	

live_trades

Column	Type	Null	Notes
id	str36	no	PK
deployment_id	str36	no	FK deployments.id, indexed
deployed_strategy_id	str36	no	FK deployed_strategies.id
open_intent_id	str36	no	FK trade_intents.id
close_intent_id	str36	yes	FK trade_intents.id
strategy_version_id	int	no	
legs	json	no	
opened_at	ts	no	
closed_at	ts	no	
holding_minutes	int	no	
exit_rule	text	yes	
exit_reason	str64	no	Includes kill_switch and broker_stop
gross	money	no	
est_costs	money	no	
actual_costs	money	yes	
net	money	no	
slippage_vs_expected	money	yes	
pnl_vs_target	money	yes	
max_adverse_excursion	money	yes	
max_favorable_excursion	money	yes	
outcome_label	str16	no	
partial	bool	no	0004. Server default false

broker_charges

Column	Type	Null	Notes
id	int	no	PK
broker_account_id	str36	no	FK broker_accounts.id, indexed
trade_date	str10	no	
order_id	str36	yes	FK orders.id
source	str32	no	
breakdown	json	no	
total	money	no	
raw	json	no	

position_snapshots

deployment_id has no foreign key. The model comment says SQLite cannot add that key without rebuilding the append-only triggers.

Column	Type	Null	Notes
id	int	no	PK
broker_account_id	str36	no	FK broker_accounts.id, indexed
deployment_id	str36	yes	Indexed. No FK
taken_at	ts	no	
positions	json	no	
mtm	money	no	

capital_events

Column	Type	Null	Notes
id	int	no	PK
deployment_id	str36	no	FK deployments.id, indexed
kind	str16	no	deposit or withdrawal
amount_inr	money	no	Always positive
note	text	yes	
actor	str128	no	
created_at	ts	no	

drawdown_events

Column	Type	Null	Notes
id	int	no	PK
deployment_id	str36	no	FK deployments.id, indexed
level	int	no	0 is normal
action	str24	no	normal, warn, reduce_size, stop_new_entries, pause
at_pct	float	yes	
drawdown_pct	float	no	
drawdown_inr	money	no	
equity	money	no	
peak	money	no	
detail	text	yes	
created_at	ts	no	

live_trade_reviews

Column	Type	Null	Notes
id	int	no	PK
live_trade_id	str36	no	FK live_trades.id, indexed
reviewer_type	str16	no	
reviewer	str128	no	llm:<model> for an agent lesson

Column	Type	Null	Notes
followed_plan	bool	yes	
process_verdict	str48	yes	
what_went_well	text	yes	
what_went_wrong	text	yes	
mistake_tags	json	no	
lesson	text	yes	
improvement_suggestion	text	yes	
suggested_rule_change	json	yes	
confidence	float	yes	
supersedes_review_id	int	yes	FK live_trade_reviews.id
created_at	ts	no	

trade_opinions

Column	Type	Null	Notes
id	str36	no	PK
deployment_id	str36	no	FK deployments.id, indexed
trade_intent_id	str36	yes	FK trade_intents.id, indexed
mode	str16	no	shadow or veto
status	str16	no	ok or error
decision	str8	yes	take, reduce, or skip
lots	int	yes	
rules_lots	int	no	
final_lots	int	no	
applied	bool	no	True only when the opinion changed the order
reasons	json	no	
risk_flags	json	no	
confidence	float	yes	
model	str100	yes	
prompt_sha256	str64	no	
context	json	no	
lesson_review_ids	json	no	
error	text	yes	
latency_ms	int	yes	
created_at	ts	no	

phantom_trades

trade_intent_id is unique.

Column	Type	Null	Notes
id	str36	no	PK
deployment_id	str36	no	FK deployments.id, indexed
trade_intent_id	str36	no	FK trade_intents.id, unique
trade_opinion_id	str36	no	FK trade_opinions.id
kind	str16	no	vetoed or reduced
lots	int	no	
legs	json	no	
opened_at	ts	no	
closed_at	ts	no	
exit_reason	str64	no	
gross	money	no	
est_costs	money	no	
net	money	no	
created_at	ts	no	

broker_credentials

Column	Type	Null	Notes
id	int	no	PK
broker_account_id	str36	no	FK broker_accounts.id, indexed
field	str64	no	
ciphertext	text	yes	Fernet. Null when revoked
fingerprint	str16	yes	12-character sha256 prefix in API responses
revoked	bool	no	Server default false
created_by	str128	no	
created_at	ts	no	Latest row per field wins

crew_heartbeats

Column	Type	Null	Notes
id	int	no	PK
deployment_id	str36	no	FK deployments.id, indexed
at	ts	no	
feed	str16	no	replay or broker
outcome	str32	yes	
pid	int	yes	

Market columns

index_bars PK (underlying, date). Columns: close, vix, iv_rank, open, high, low (all float, null allowed except the key).

option_eod PK (underlying, date, expiry, strike, option_type). Other columns: close, settle, volume (int, default 0), open_interest, lot_size, underlying_price.

import_files PK sha256 str64. Columns: name str300, kind str32, rows int, first_date, last_date, imported_at.

market_events unique (date, kind, title). Columns: id, date (indexed), kind, title, underlying, source.

news_items: id, source, title, url (unique, str1000), published_at (indexed), fetched_at, symbol.

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4. Relationships

Research, from child to parent:

- strategy_versions.strategy_id → strategies
- strategy_versions.ingest_run_id → ingest_runs
- reports.strategy_id → strategies
- reports.strategy_version_id → strategy_versions
- simulation_runs.report_id → reports
- sim_trades.simulation_run_id → simulation_runs
- sim_trades.report_id → reports
- sim_trades.strategy_version_id → strategy_versions
- trade_reviews.sim_trade_id → sim_trades
- trade_reviews.supersedes_review_id → trade_reviews
- strategy_improvements.strategy_id → strategies
- strategy_improvements.from_version_id → strategy_versions
- strategy_improvements.report_id → reports
- improvement_decisions.improvement_id → strategy_improvements
- improvement_decisions.resulting_version_id → strategy_versions
- paper_orders.strategy_id → strategies

Live, from child to parent:

- deployments.deployed_strategy_id → deployed_strategies
- deployments.broker_account_id → broker_accounts
- deployment_events.deployment_id → deployments
- trade_intents.deployment_id → deployments
- trade_intents.deployed_strategy_id → deployed_strategies
- orders.trade_intent_id → trade_intents
- order_events.order_id → orders
- fills.order_id → orders
- live_trades.deployment_id → deployments
- live_trades.deployed_strategy_id → deployed_strategies
- live_trades.open_intent_id and close_intent_id → trade_intents
- broker_charges.broker_account_id → broker_accounts
- broker_charges.order_id → orders
- position_snapshots.broker_account_id → broker_accounts
- capital_events.deployment_id → deployments
- drawdown_events.deployment_id → deployments
- live_trade_reviews.live_trade_id → live_trades
- live_trade_reviews.supersedes_review_id → live_trade_reviews
- trade_opinions.deployment_id → deployments
- trade_opinions.trade_intent_id → trade_intents
- phantom_trades → deployments, trade_intents, trade_opinions
- broker_credentials.broker_account_id → broker_accounts
- crew_heartbeats.deployment_id → deployments
- live_audit_events has no foreign keys

`position_snapshots.deployment_id` and every `strategy_id` / `strategy_version_id` / `report_id` column on a live table are strings with no foreign key.

Market tables have no foreign keys.

---

5. Identifiers that cross databases

Value	Written in research	Copied into live
`strategies.id`	PK	`deployed_strategies.strategy_id`, `deployments.strategy_id`
`strategy_versions.id`	PK	`deployed_strategies.strategy_version_id`, `deployments.strategy_version_id`, intents and live trades
`reports.id`	PK	`deployed_strategies.report_id`
Content hash	`strategy_versions.content_sha256`	`deployed_strategies.content_sha256`
Rules hash	Inside version frontmatter; recomputed at approval	`deployed_strategies.rules_sha256`

The Trade Crew recomputes both hashes from the frozen row and halts on a mismatch.

---

6. Data lifecycle

- Inserts are the normal write. Append-only tables reject ORM `UPDATE` and `DELETE` and the matching SQL through triggers installed by each migration (`backend/app/db/triggers.py`).
- `strategies` allows update and rejects delete. Archive sets `archived_at`.
- A correction to a review is a new row with `supersedes_review_id`.
- An applied improvement inserts `improvement_decisions` and a new `strategy_versions` row. It does not edit the old version.
- Credential rotation inserts a new `broker_credentials` row. Revocation inserts a row with `revoked` true and no ciphertext.
- Market rows may be updated. `import_files.sha256` skips a file that was already imported.
- `trade_journal` is a view over simulated trades. Live trades are read from the live database by `JournalStore` and merged in Python.

---

7. Database boundaries

Data	Where it lives
Strategy text and review history	Research
Manual paper orders	Research (`paper_orders`). Not the live `orders` table
Delayed quotes and candles	Not stored. Fetched from Yahoo per request
Orders, fills, positions, credentials	Live
NSE closes, events, headlines	Market
Original video or HTML bytes	Not stored. Extracted text is on `ingest_runs.source_text`
Sandbox working state	JSON files under `SANDBOX_STATE_DIR`, not a table
API tokens	`API_TOKENS` hashes in the environment, not a table

---

8. Mermaid ER diagram

```
erDiagram
  strategies ||--o{ strategy_versions : has
  ingest_runs ||--o{ strategy_versions : produced
  strategies ||--o{ reports : reviewed
  strategy_versions ||--o{ reports : of
  reports ||--o{ simulation_runs : contains
  simulation_runs ||--o{ sim_trades : contains
  sim_trades ||--o{ trade_reviews : reviewed
  strategies ||--o{ strategy_improvements : proposes
  strategy_improvements ||--o{ improvement_decisions : decided
  strategies |o--o{ paper_orders : tags
```

```
erDiagram
  deployed_strategies ||--o{ deployments : freezes
  broker_accounts ||--o{ deployments : hosts
  deployments ||--o{ deployment_events : status
  deployments ||--o{ trade_intents : decides
  trade_intents ||--o{ orders : sends
  orders ||--o{ order_events : status
  orders ||--o{ fills : fills
  deployments ||--o{ live_trades : books
  deployments ||--o{ capital_events : cash
  deployments ||--o{ drawdown_events : ladder
  deployments ||--o{ trade_opinions : opinion
  trade_opinions ||--o{ phantom_trades : counterfactual
  broker_accounts ||--o{ broker_credentials : secrets
  deployments ||--o{ crew_heartbeats : ticks
```

---

9. Source evidence

Claim	File
Research models	`backend/app/db/models/core.py`, `sim.py`, `journal.py`
Live models	`backend/app/db/models/live.py`
Market models	`backend/app/market/models.py`
Append-only and no-delete	`backend/app/db/base.py`, `guards.py`, `triggers.py`
Research head	`backend/alembic/versions/0006_paper_orders.py`
Live split	`backend/alembic/versions/0004_split_live.py`
Live head	`backend/alembic_live/versions/0005_broker_credentials.py`
Money and timestamps	`backend/app/db/types.py`
View	`backend/alembic/versions/0004_split_live.py` (`SIM_JOURNAL_VIEW`)